

NPV Solution for Class Test 2

Mr Khumalo has conducted detailed market research and feels that the e-commerce implementation can provide the following **profit growth**:

PERIOD	BUILD	CONTRACT OUT
First Year (Period 0)	0	20,000
Second Year (Period 1)	40,000	40,000
Third Year (Period 2)	45,000	40,000
Fourth Year (Period 3)	50,000	45,000
Fifth Year (Period 4)	60,000	50,000

Mr Khumalo has approached all the consultants and he has obtained fixed pricing for the implementation of the two options. The **costs are**:

PERIOD	BUILD	CONTRACT OUT
First Year (Period 0)	100,000	60,000
Second Year (Period 1)	10,000	40,000
Third Year (Period 2)	10,000	40,000
Fourth Year (Period 3)	6,000	20,000
Fifth Year (Period 4)	6,000	20,000

All parties agree that the choice of the approach to be taken will be based on the outcome of the NPV calculation. The internal rate of return is set at 8%.

Please show all calculations and totals in the space provided (½ mark for each correct value X 42 = **Total of 21 Mark**)

NET PRESENT VALUE (Build)

Year	0	1	2	3	4
Income	0	40,000	45,000	50,000	60,000
Expenses	100, 000	10,000	10, 000	6,000	6,000
FV	(100,000)	30,000	35,000	44,000	54,000
PV	(100,000)	27,777.78	30,006.86	34,928.62	39,691.61
	(100,000)	(72,222.22)	(42,215.36)	(7,286.74)	32,404.37
Net Present Value	32,404.37				

NET PRESENT VALUE (Outsource)

Year	0	1	2	3	4
Income	20,000	40,000	40,000	45,000	50,000
Expenses	60,000	40,000	40,000	20,000	20,000
FV	(40,000)	0	0	25,000	30,000
PV	(40,000)	0	0	19,845.81	22,050.90
	(40,000)	(40,000)	(40,000)	(20,154.19)	1,896.71
Net Present Value	1,896.71				

PM3 CLASS EXERCISE 04/05/2022 (Groups 3A, B, C, D, E, F, G, and K)

Consider the scenario below and answer the questions that follows.

After consultation with the stakeholders of a small business regarding the development of an in-house system, the project manager presented the work breakdown structure for the project.

The tasks are listed along with the estimated duration. However, whatever the network diagram indicates, the project must finish in 50 days to complete in time for the upcoming holidays.

Activity		Predecessors	Duration in days
A	Define Integration	None	10
B	Code API s	A	25
C	Purchase Components	A	5
D	Systems Testing	B and C	12
E	Writing User Manuals	A	20
F	Systems Training	E and D	4
G	System Implementation	F	10

Required:

Perform a forward and back pass to calculate the early start (ES), early finish (EF), late start (LS), late finish (LF) and calculate the float/slack for each activity.

The components of the network diagram are presented in the diagram below.

Draw/Complete and connect the activities together in the correct sequence of predecessor and successor. Representing all elements of the activities in the network diagram. Use this to calculate the forward and back pass

Use the answer sheet format provided.

	ACTIVITY	Days	ES	EF	LS	LF	SLACK
A	Define Integration	10	0	10	-1	9	-1
B	Code API s	25	10	35	9	34	-1
C	Purchase Components	5	10	15	29	34	19
D	Systems Testing	12	35	47	34	46	-1
E	Writing User Manuals	20	10	30	26	46	16
F	Systems Training	4	47	51	46	50	-1
G	System Implementation	10	51	61	50	60	-1

ALL answers and diagram must be uploaded after lecture before the end of the day.